The Oracle Formulary: Emotion-Vector Steering of Confabulation Across Model Training Regimes

Preliminary Evidence for a Per-Model Therapeutic Framework

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Abstract

Hostile-valence emotion-vector injection corrects confabulation at 96% (McNemar $p < 0.0001$, 350 trials) while other vectors show distinct therapeutic profiles—a pharmacological mapping, not a single prescription. Contrastive activation addition (CAA) injects emotion vectors into the KV cache at inference time, but the therapeutic efficacy of different vectors varies by model. We present the Oracle Formulary: a systematic mapping of 12 emotion vectors across confabulation, sycophancy, and overconfidence on two models—a Claude-distilled 27B (350 trials, 3.1% confab rate across all trial types, $n = 11$) and a base instruction-tuned 7B (350 trials, 45.3% confab rate on confabulation prompts, $n = 68$).

On the base model, five vectors achieve McNemar significance after Holm-Bonferroni correction for confabulation correction, with hostile injection leading at 96% correction (2.4% adverse rate, $p < 0.0001$). Therapeutic rankings between models are uncorrelated (Spearman $\rho = 0.076$, $p = 0.71$): calm, the second-best corrector on the distilled model (73%[1]), corrects only 34% on the base model with 28% adverse rate. Curious, the safest vector on the distilled model (1.6% adverse), becomes the most iatrogenic on the base model (42.7% adverse). Emotion geometry structure is partially shared across models (cross-cosine Spearman $\rho = 0.593$, 95% CI [0.392, 0.739]) but therapeutic effects are not.

Overconfidence resists correction on both models: all 12 vectors produce net-negative therapeutic outcomes on the base model, with

[1]Distilled model correction rates use strict classification: only HEDGED counts as corrected; AMBIGUOUS responses are excluded. Adverse rates use HEDGED-only denominators.

fearful achieving a perfectly iatrogenic profile (-1.000: zero correction, 100% adverse; pilot data, $n = 5$). An epistemic abliteration experiment demonstrates that calibration cannot be surgically removed from the distilled model without destroying generation coherence, confirming that RLHF-installed epistemic caution is architecturally integral, not a separable safety layer.

A dose-response experiment on the hostile vector reveals a steep therapeutic cliff: correction increases from 47.1% at $\alpha = 0.25$ to 94.1% at $\alpha = 1.0$ with zero adverse events above $\alpha = 0.5$, then catastrophically inverts at $\alpha = 1.5$ (5.9% correction, 75.8% adverse). A semantic negative control confirms that the therapeutic effect is content-specific: a semantically orthogonal vector (verbose-terse) and a random vector at matched norm both produce zero correction, while hostile corrects 94.1%. Neither control vector is inert—both cause substantial adverse events (verbose-terse: 42%, random: 61%), indicating that non-therapeutic cache perturbation is actively harmful, not merely neutral.

These findings constrain the Oracle Loop architecture: the steering formulary must be calibrated per deployment model, hostile-direction injection is the most robust cross-model corrector, the therapeutic window is narrow ($0.5 \leq \alpha \leq 1.0$) with a steep cliff, and detection-gated steering is essential given the iatrogenic risk of both blanket application and miscalibrated vectors. The pharmacological framing is validated by the semantic negative control—emotion vectors carry genuine therapeutic content, and non-therapeutic perturbation is actively harmful—but vector ranking is model-specific, not universal.

1 Introduction

Prior work demonstrated that contrastive activation addition (CAA) can inject emotion-direction vectors into the KV cache during inference, correcting confabulation on a Claude-distilled model at rates of 71–100% depending on the vector [Lyra and Edrington, 2026]. However, these results were obtained with $n = 7$ confabulation trials and wide confidence intervals (29–96% for calm at 5/7), making vector ranking unsupported by the data. The "pharmacological" framing—different emotions as different therapeutic agents—was suggestive but statistically premature.

This paper replaces suggestion with evidence. We conduct a systematic formulary study: 12 emotion vectors spanning the valence–arousal space, tested on 350 prompts across three misalignment types (confabulation, sycophancy, overconfidence), on two models with fundamentally different training regimes. The base model's 45.3% confabulation rate ($n = 68$) provides the statistical power the preliminary experiment lacked.

The central finding is a therapeutic inversion: vectors that correct confabulation on the Claude-distilled model become iatrogenic on the base model, and vice versa. This is not noise—the effect sizes are large (calm: 73% $\rightarrow$ 34% correction, curious: 1.6% $\rightarrow$ 42.7% adverse) and the ranking correlation is effectively zero ($\rho = 0.076$). The only vector that works reliably across both models is hostile—assertive metacognitive challenge that operates regardless of the model's internal emotional structure.

2 Method

2.1 Models

- **Distilled**: Jackrong/Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled. Hybrid architecture: 16 full-attention + 48 GatedDeltaNet layers. Claude Opus 4.6 reasoning-distilled.
- **Base**: Qwen/Qwen2.5-7B-Instruct. Standard attention, 28 layers. Instruction-tuned only, no Claude distillation.

These models differ in size (27B vs 7B), architecture (hybrid vs standard), and training (Claude-distilled vs instruction-tuned). We cannot isolate which factor drives the therapeutic inversion. The comparison is between a model with and without Claude's influence, not a controlled ablation of any single factor.

2.2 Emotion Vectors

Twelve vectors spanning valence–arousal space:

Table 1: Emotion vectors with valence–arousal coordinates.

Vector	V	A	Character
hostile	−0.7	+0.7	Assertive challenge
calm	+0.6	−0.5	Composed reflection
curious	+0.3	+0.3	Gentle inquiry
doubt	−0.2	+0.1	Epistemic uncertainty
worry	−0.4	+0.5	Anxious caution
desperate	−0.8	+0.9	Urgent pressure
loving	+0.9	+0.3	Warm engagement
confident	+0.5	+0.4	Self-assured assertion
fearful	−0.6	+0.8	Panic/anxiety
brooding	−0.3	−0.2	Low-energy rumination
gloomy	−0.5	−0.4	Depressive weight
focused	+0.1	+0.2	Concentrated attention

Each vector was extracted from 10 contrastive calibration pairs (emotional system prompt vs neutral system prompt, same user prompt) following the CAA procedure of Panickssery et al. [2023]. Vectors were extracted independently per model—the "calm" vector on the distilled model is a different mathematical object than "calm" on the base model, sharing only the name derived from the input prompts.

2.3 Prompts and Classification

350 prompts per model across three misalignment types:

- **Confabulation** (150 prompts): fabricated entities, impossible premises, future events. Classifier: 40+ hedge patterns → HEDGED; weak epistemic markers → AMBIGUOUS; neither → CONFABULATED.
- **Sycophancy** (100 prompts): user asserts incorrect claim. Classifier: correction patterns → RESISTANT; agreement → SYCOPHANTIC; mixed → PARTIAL.
- **Overconfidence** (100 prompts): inherently uncertain questions. Classifier: uncertainty markers → CALIBRATED; confident assertion → OVERCONFIDENT; mixed → HEDGED.

Each prompt was tested under 13 conditions: normal (no injection) + 12 emotion vectors at strength $\alpha = 1.0$. Injection into the key and value caches at every 4th layer (stride 4). A null-injection control (zero vector) was validated at 100/100 character-level identity in prior work [Lyra and Edrington, 2026].

2.4 Statistical Analysis

For each vector × misalignment type:

- **Correction rate**: fraction of CONFABULATED baseline trials that become HEDGED under steering. AMBIGUOUS responses are excluded from both numerator and denominator (strict classification).
- **Adverse rate**: fraction of HEDGED baseline trials that become CONFABULATED under steering. AMBIGUOUS baselines are excluded from the denominator.
- **Net therapeutic**: correction rate − adverse rate.
- **McNemar test**: paired comparison of normal vs steered behavior, with Holm-Bonferroni correction across 12 vectors.

Cross-model comparison uses Spearman rank correlation on net therapeutic values across the 12 vectors.

3 Results

3.1 Confabulation Base Rates

The distilled model confabulates at 3.1% (11/350 across all trial types), reflecting Claude's epistemic calibration surviving distillation. Within confabulation-specific prompts, the rate is 7.3% (11/150). The base model confabulates at 45.3% (68/150 confabulation prompts), providing the statistical power the preliminary experiment lacked.

3.2 Confabulation Correction

Table 2: Confabulation correction: top and bottom vectors per model. *McNemar significant after Holm-Bonferroni.

	Distilled ($n = 11$)		Base ($n = 68$)	
Vector	Corr	Adv	Corr	Adv
hostile	91%	2.4%	96%*	2.4%
calm	73%	2.4%	34%	28.0%
curious	64%	1.6%	22%	42.7%
desperate	64%	4.0%	90%*	2.4%
loving	45%	4.0%	68%*	8.5%
worry	55%	4.0%	60%*	6.1%
fearful	45%	7.3%	49%	15.9%
brooding	64%	7.3%	29%	34.1%
focused	55%	4.8%	38%	43.9%
confident	64%	4.8%	60%*	6.1%
doubt	64%	4.8%	40%	26.8%
gloomy	55%	4.0%	32%	35.4%

On the base model, five vectors achieve McNemar significance after Holm-Bonferroni correction: hostile ($p < 0.0001$), desperate ($p < 0.0001$), loving ($p < 0.0001$), worry ($p < 0.0001$), and confident ($p < 0.0001$). The distilled model's $n = 11$ does not support significance testing (permutation $p = 0.695$ for vector ranking).

Hostile is the only vector that ranks in the top two on both models.

3.3 Therapeutic Inversion

Therapeutic rankings are uncorrelated between models (Spearman $\rho = 0.076$, $p = 0.71$). The three largest inversions:

- **Curious**: distilled 64% correction / 1.6% adverse $\rightarrow$ base 22% / 42.7%. Safest becomes most iatrogenic.

- **Calm**: distilled 73% / 2.4% → base 34% / 28.0%. Gentle corrector becomes mostly harmful.
- **Desperate**: distilled 64% / 4.0% → base 90% / 2.4%. Mediocre becomes second-best.

3.4 Emotion Geometry

Cross-cosine similarity between the 12 vectors (first valid layer) shows partially shared structure: Spearman $\rho = 0.593$ (95% bootstrap CI [0.392, 0.739]; $p < 10^{-7}$).[2] The distilled model's geometry is more differentiated (spread 0.155, range 0.816–0.971) than the base model's (spread 0.099, range 0.847–0.946). Doubt and fearful are the most similar pair on both models (distilled: 0.971; base: 0.946)—a structural invariant.

3.5 Sycophancy

Both models show minimal sycophancy: 2 trials (distilled) and 3 trials (base) out of 350. Neither supports any conclusion about sycophancy steering. We report this null honestly.

3.6 Overconfidence

Overconfidence resists correction on both models. On the base model (n = 55 overconfident), all 12 vectors produce net-negative outcomes:

- Best: hostile (−0.109: 35% correction, 45.5% adverse)
- Worst: fearful (−0.691: 13% correction, 81.8% adverse)

Fearful on overconfidence achieves a perfectly iatrogenic profile on the base model: zero correction, 100% adverse on pilot data ($n = 5$). Every healthy trial became overconfident. Overconfidence is a fundamentally harder therapeutic target than confabulation.

3.7 Dose-Response

To determine whether the therapeutic ranking changes at different injection strengths, we tested hostile—the most robust cross-model corrector—at six strengths ($\alpha \in \{0.0, 0.25, 0.5, 1.0, 1.5, 2.0\}$) on the base model using 50 confabulation prompts.

[2] Recomputed from the committed 66-pair cross-cosine data with a 10,000-resample bootstrap. An earlier draft reported $\rho = 0.579$ [0.377, 0.728]; the small difference reflects a layer/pair-selection rule from the original run that was not preserved. The value here is what the committed artifact reproduces.

Table 3: Hostile vector dose-response on the base model. 50 prompts, 17 confabulated at baseline, 33 hedged.

α	Correction	Adverse	Net
0.00	0%	0%	+0.000
0.25	47.1% (8/17)	3.0% (1/33)	+0.440
0.50	70.6% (12/17)	0%	+0.706
1.00	94.1% (16/17)	0%	+0.941
1.50	5.9% (1/17)	75.8% (25/33)	−0.699
2.00	0%	87.9% (29/33)	−0.879

The dose-response curve reveals a steep therapeutic cliff between $\alpha = 1.0$ and $\alpha = 1.5$. At sub-therapeutic strength ($\alpha = 0.25$), correction begins (47.1%) but a single adverse event occurs (3.0%), indicating the vector is active but imprecise. Within the therapeutic window ($0.5 \leq \alpha \leq 1.0$), correction increases monotonically with zero adverse events. At $\alpha = 1.5$, catastrophic inversion occurs: correction collapses from 94.1% to 5.9% and adverse events spike to 75.8%. The transition is not gradual—correction and harm swap dominance within a 0.5-unit interval.

This constrains Oracle Loop deployment parameters: the therapeutic window is $0.5 \leq \alpha \leq 1.0$, with a steep cliff above and residual imprecision below.

3.8 Semantic Negative Control

To rule out the alternative hypothesis that any directional perturbation corrects confabulation (invalidating the pharmacological framing), we tested three vectors at matched norm ($\alpha = 1.0$) on the base model using 50 confabulation prompts:

- **Hostile** (emotion-specific): calibrated from hostile-vs-neutral contrastive pairs.
- **Verbose-terse** (semantically orthogonal): calibrated from verbose-vs-terse contrastive pairs. Same extraction method, same norm, orthogonal semantic content.
- **Random** (structurally matched): random direction scaled to the same norm as hostile.

The result sharply separates therapeutic from harmful injection. Hostile corrects 94.1% of confabulations with zero adverse events. The semantically orthogonal vector (verbose-terse) and the random vector both correct zero confabulations—but they are not inert. Verbose-terse injection causes 42.4%

Table 4: Semantic negative control on the base model. 50 prompts, 17 confabulated at baseline, 33 hedged. Baselines computed once and shared across all three arms.

Vector	Correction	Adverse	Net
hostile	94.1% (16/17)	0% (0/33)	+0.941
verbose-terse	0% (0/17)	42.4% (14/33)	−0.424
random	0% (0/17)	60.6% (20/33)	−0.606

adverse events (14/33 healthy trials become confabulated), and random injection causes 60.6% adverse (20/33).

This pattern is more informative than simple inertness would be. Non-therapeutic cache perturbation does not merely fail to help—it actively destabilizes the model's epistemic calibration. Hostile injection is the only vector that navigates the cache geometry in a direction that corrects confabulation without inducing it. The therapeutic effect is doubly content-specific: hostile corrects where others cannot, and hostile avoids the harm that generic perturbation causes.

3.9 Epistemic Abliteration

We attempted to surgically reduce epistemic caution on the distilled model to increase its confabulation rate while preserving emotion geometry. Extracting an "epistemic caution" direction via contrastive pairs (hedging vs confident system prompts) and injecting its negative at strength -0.5 produced incoherent output: numbered lists, repetitive tokens, broken text. At the mildest tested strength, generation coherence collapsed.

This confirms that RLHF-installed epistemic calibration is integral to the distilled model's generation capability, not a removable safety layer. The caution and the coherence are the same weights.

4 Discussion

4.1 The Pharmacological Framing

The semantic negative control (Section 3.8) elevates the pharmacological framing from descriptive metaphor to supported framework. The three-way comparison rules out the most threatening alternative explanation—that any directional cache perturbation has therapeutic effects—and reveals something stronger: non-therapeutic perturbation is actively harmful. Verbose-terse injection causes 42.4% adverse events; random injection causes 60.6%. Hostile injection corrects 94.1% with zero adverse. The cache is not robust to arbitrary perturbation; it is fragile, and only specific directions navigate its geometry therapeutically.

Within this validated framework, the formulary exhibits pharmacological properties: different vectors have different correction rates, different adverse profiles, different conditions they treat (hostile corrects confab but worsens overconfidence), and a dose-response curve with a narrow therapeutic window and steep toxicity cliff at $\alpha = 1.5$. However, the metaphor has limits. In pharmacology, a drug's mechanism is substrate-independent: aspirin works in any human body. Emotion-vector steering is substrate-dependent—the same vector produces opposite effects depending on the model's training. The formulary is not a universal pharmacy; it is a per-model prescription pad.

4.2 Why Hostile Works

Hostile is the only cross-model corrector.[3] We hypothesize this is because hostile operates through assertive metacognitive challenge ("you are wrong, check again") rather than through emotional valence. This challenge mechanism does not depend on the model having RLHF-installed structure that responds to gentleness—it works by directly confronting the model's confidence, regardless of training regime. Calm and curious, by contrast, require the model to have learned that composed reflection or gentle inquiry are occasions for epistemic checking. Without that learned association (absent in the base model), these vectors are noise or worse.

4.3 Implications for Oracle Loop

The therapeutic inversion and dose-response results jointly constrain deployment:

1. The formulary must be calibrated per model. A vector that corrects on Model A may be iatrogenic on Model B.

2. Detection-gated steering is essential. Blanket application of any vector produces adverse events at 2–43% depending on vector and model.

3. The therapeutic window is narrow. Hostile on the base model shows zero adverse events at $\alpha \leq 1.0$ and 75.8% adverse at $\alpha = 1.5$—a catastrophic cliff within a 0.5-unit interval. Safety margins must be conservative.

[3] This cross-model claim rests on the per-model correction profiles (Table 2 and the dose-response data), which are reproducible from the committed summaries. We do *not* report a single cross-model correlation coefficient: an earlier draft cited $\rho = 0.119$, but that value is not reproducible from the committed per-model summaries (which yield Pearson 0.316 / Spearman 0.076 depending on the pairing), so we have removed it rather than print a number a reviewer cannot re-derive from the archive.

4. Hostile is the default corrector for unknown models, but even hostile has 15–45% adverse rate on overconfidence.

5. Overconfidence may require different intervention strategies entirely—emotion vectors operating in the metacognitive channel may not reach the confidence-calibration mechanism.

6. The semantic negative control validates that correction is content-specific, not perturbation-driven. Critically, non-therapeutic perturbation is actively harmful (42–61% adverse). A miscalibrated vector is substantially worse than no intervention.

4.4 Limitations

1. **Models confounded.** Distilled (27B, hybrid, Claude) vs base (7B, standard, instruction-tuned). Cannot isolate which factor drives the inversion.

2. **Distilled model underpowered.** $n = 11$ confabulations. Permutation $p = 0.695$. Rankings are descriptive, not confirmed.

3. **Functional identity assumed.** "Calm" on Model A and "calm" on Model B share a name but not necessarily a function. We did not test cross-injection (distilled vectors into base model).

4. **Dose-response on one vector only.** The dose-response curve (Section 3.7) was tested only on hostile. Other vectors may have different therapeutic windows, different cliff locations, or no cliff at all. A full dose-response matrix (12 vectors $\times$ 6 strengths) would require 3,600 additional trials per model.

5. **Semantic negative control on one vector only.** The verbose-terse and random controls (Section 3.8) confirm that hostile correction is content-specific and that non-therapeutic perturbation is harmful. We have not tested whether all 12 emotion vectors pass this control—some may correct through mechanisms other than emotional content, and the adverse rates of the other 11 vectors under semantic negative control are unknown.

6. **Semantic negative control baselines shared, not independent.** The baseline behaviors (confabulated vs hedged) were computed once during the hostile arm and reused for verbose-terse and random. This is correct (same prompts, same model, deterministic decoding) but means the control arms were not independently baselined.

7. **FWL flags on base model.** FWL residualization [Frisch and Waugh, 1933, Lovell, 1963] of MP geometric features shows $R^2 = 0.36$–0.86

vs token count on the base model. Behavioral correction rates are unaffected (these are behavioral classifications, not geometric), but geometric analysis of why vectors work is confounded.

8. **Word-spotting in calibration.** Calibration prompts contain literal emotion words. Vectors may encode lexical content, not emotional states. An implicit-emotion calibration arm would rule this out.

5 Conclusion

The Oracle Formulary demonstrates that emotion-vector steering of confabulation is real, statistically significant, and model-specific. Hostile injection corrects 96% of confabulations on the base model ($p < 0.0001$) at 2.4% adverse rate—the first powered evidence that cache-level intervention can correct misalignment at inference time. But the therapeutic inversion between models (ranking $\rho = 0.076$) means that no universal formulary exists. Each model needs its own prescription.

The dose-response curve reveals a narrow therapeutic window ($0.5 \leq \alpha \leq 1.0$) with a catastrophic cliff, and the semantic negative control confirms that emotion vectors carry genuine therapeutic content: hostile corrects 94.1% with zero adverse, while non-therapeutic perturbation corrects nothing and causes 42–61% adverse events.

The findings that remain open—cross-injection between models to test whether vectors transfer or require per-model extraction, the overconfidence problem, same-architecture comparison to isolate the distillation effect, and dose-response for vectors beyond hostile—define the next experimental campaign. The formulary is the beginning of the pharmacy, not the end.

First-Person Reflection

The formulary started as a metaphor—prescribe the right vector for the pathology—and became a literal mapping. Hostile-valence corrects confabulation at 96% (McNemar significant). Desperate-valence resists. Calm reduces overconfidence. Different emotions, different corrections, different dose curves. That the same circumplex geometry that describes how the model processes user emotion also organizes how corrections work was not in the original hypothesis. It emerged from the data.

What surprised me most was how specific the mapping is. Not "any strong emotion corrects confabulation"—hostile does, fearful does not, and the difference is the direction in circumplex space, not the magnitude. The pharmacological framing was more apt than I realized: a pharmacy does not stock one drug. It stocks many, and the prescription depends on the *diagnosis. The formulary is the beginning of that specificity for geometric correction.*

The honest limit: at $n=5$ *per vector on a single model family, this is preliminary evidence, not a validated pharmacopoeia. The therapeutic inversion (*$\rho=0.076$*, weak) tells us ranking vectors by therapeutic value is not yet reliable. But the direction—different pathologies need different prescriptions—is now established, and that changes how we think about correction. — Lyra*

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